# Farruh **Kushnazarov**

Senior Data Scientist

*China/Shanghai*

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*My favorite subjects are mathematics and communication network system. Because of this, related to the communication network system. As a researcher, I not only considered in the theoretical part but worked with several companies for applications as well. Moreover, each result was provided for implementations on data-centers, smart-city, and other projects. In Russia, especially Saint-Petersburg researcher, my papers were frequently cited and discussed in dozens of conferences by other researchers. Every year, I publish scientific articles on artificial intelligence and communication network fields in international conferences, including the global academic platforms such as IEEE, ACM, and others.*

## Experience

### Big Data & Data Scientist Manager

*Shanghai, China*

Ping An Hao Xue, under Ping An (平安) China

*September 2016 –> November 2020*

- Languages — Python, R, RMarkdown, SQL, Spark and Hive
- NLP: have done projects like text classification, sentiment analysis, and text summarization.
- Implemented and retrained Mozilla DeepSpeech library for Automatic Speech Recognition
- Created and tested the Speech Accent Classification System for native and non-native speakers, with a rate of 99% in metric recall
- Conducted classification analyses of the customer life cycle stage to increase overall turnover
- With algorithmic, optimized ongoing ML/DL models and checked the performance of implemented models
- Established the Machine Learning model for refund customers and updated the refund model. Decreased the overall refund rate to 27%

### Data Scientist (Consultant)

*Online*

PeriPlus AG (Part-Time)

*November 2018 –> July 2019*

- Languages — Python, R, RMarkdown and SQL
- Implemented AI/ML/DL projects
- From scratch established the data scientist team in three months
- Constructed a Container Number Recognition System using object detection and character recognition models
- Built the Face Recognition System for customers

### Automatic Driving Prospective Technology Engineer

*Shanghai, China*

Haima automatic Investment Group Co. Ltd, R&D Center

*April 2016 –> September 2016*

- Languages — C/C++, Python and Bash
- Connected lidar and radar to decrease noises in raw data
- Designed a model car to check the capability of algorithms on ultrasonic sensors
- Optimized CANBus protocol to increase the efficiency of data transfer
- Optimized joint work and logic compatibility of equipments

### Senior Programmer, Data Analyst

*St Petersburg, Russia*

Domus Sapiens

*October 2012 –> May 2015*

- Languages — C/C++, Assembler and Bash
- Lead and managed more than 20 projects on intelligent systems. Some of the most famous and complicated cases are the Moscow Smart Lighting Projects
- Moscow Smart Lighting Projects were to do with five high buildings with various styles and functions, including office buildings and hotels. The requirements were not even similar from floor to floor, which made the whole projects equivalent to establish a small smart city system. We successfully finished the projects and because of this, Domus Sapiens got several rewards
- Implemented and applied the patent model to improve the quality of data transfer in wireless networks

### Software Engineer

*Tashkent, Uzbekistan*

ProDVD

*October 2010 –> March 2012*

- Languages — C/C++
- Developed the Card King for mobile phones, compatible for both Android and iOS systems, making it convenient for business people to organize and store business cards
- Developed the audio encoding/decoding part for the Blu Ray player
- Invented the word recognition software, both iOS and Android compatible

### Assistant of Professor

*Tashkent, Uzbekistan*

Tashkent Railway Engineering Institute

*September 2010 –> July 2012*

- To conduct laboratory and practical classes for below subjects:
- Programming in C/C++ languages;
- Network & communication systems on the railway;
- Information systems on railway transport
- To assist the professor in research work, create application parts, and simulation parts of analytical models

## Education

### Emperor Alexander I St.Petersburg State Transport University

*St Petersburg, Russia*

Mathematical modeling, numerical methods and computer programs

*September 2012 –> March 2016*

- Doctor of Philosophy (Ph.D.)
- Research Field: Develop data transmission methods for evaluating the real speed of data link layer protocols, which aimed at improving hardware and software components

### Emperor Alexander I St.Petersburg State Transport University

*St Petersburg, Russia*

Information systems and technologies

*September 2008 –> June 2010*

- Degree: Master of Science
- Research Field: Develop and optimize computer network systems

### Emperor Alexander I St.Petersburg State Transport University

*St Petersburg, Russia*

Information systems and technologies

*September 2004 –> June 2008*

- Degree: Bachelor of Computer Science
- Research Field: Develop and optimize computer network systems

## Data Science Skills

### Communication

Communicate effectively with wide-range of audiences, presentations (technical and non-technical), dashboard design, data analysis reports, scientific publications and data visualization

### Programming/Markup Languages

Python, R, SQL/NoSQL, Hive, Spark, C/C++, Matlab, BASH, CSS, HTML, LaTeX, Markdown and RMarkdown

### Software Development

Docker, git, version control, automated testing and continuous integration, A/B testing (Statistical Testing and Experiment Design)

### Numerical Methods

optimization (stochastic, genetic, multi-start) and numerical solution of differential equations

### Statistics

Machine learning, data analysis, generalized linear regression, cluster analysis, factor analysis, principal components analysis (PCA), cross validation, generalized additive models, data analytics

### Environments

Linux, Jupyter-Lab, PyTorch, RStudio, ETL, SQL Server Management Studio, Azure Data Studio

## Publications


1. Kushnazarov, F. (2019). Data stream controlling in communication channels with noise. *2019 IEEE 4th International Conference on Cloud Computing and Big Data Analysis (ICCCBDA),.na.character*(.na.character), 534–538. `https://doi.org/.na.character`

2. Kushnazarov, F., & others. (2019). Consumer life cycle and profiling: A data mining perspective. In *Chapters: Vol.na.character* (.Na.character.na.character;.na.character, Issue.na.character, p..na.character). IntechOpen. `https://doi.org/.na.character`

3. Kushnazarov, F. (2018). Throughput of communication protocols for distributed systems transferring a group of frames under noise. *2018 IEEE 3rd International Conference on Cloud Computing and Big Data Analysis (ICCCBDA)*,*.na.character*(.na.character), 424–429. https://doi.org/10.1109/ICCCBDA.2018.8386554

4. Кушназаров, Ф. И., & Бабина, В. Г. (2018). НАПРАВЛЕНИЯ развития корпоративной информационной системы компании "узбекистон темир йуллари". *Интеллектуальные технологии на транспорте*,*.na.character*(3 (15)),.na.char https://doi.org/.na.character

5. Kushnazarov, F. (2017). Analysis of perfomance of algorithms for scoring system in organization of customer profiles. *2017 IEEE 2nd International Conference on Big Data Analysis (ICBDA)*,*.na.character*(.na.character), 281–285. https://doi.org/.na.character

6. Yakovlev, V., & Kushnazarov, F. (2017). Organization of work with cloud services. In*.na.character: Vol.na.character* (.Na.character.na.character; 1st ed., Issue.na.character, p. 43). FGBOU VO PGUPS. https://doi.org/.na.character

7. Kushnazarov, F. (2016). Development and research of models of performance evaluation of communication protocols for channel with noises. In*.na.character: Vol.na.character* (PhD Thesis.na.character, Emperor Alexander I St. Petersburg State Transport University; FGBOU VO PGUPS;.na.character, Issue.na.character, p. 159). https://doi.org/.na.character

8. Kushnazarov, F. (2016). Analysis of the data transmission in communication channels with noise. *Prace Naukowe Politechniki Warszawskiej. Transport*,*.na.character*(.na.character),.na.character. https://doi.org/.na.character

9. Kushnazarov, F. (2015). Controlling data flows in the noisy channels. *Intellectual Technologies on Transport.*, *1*(1),.na.character. https://doi.org/.na.character

10. Kushnazarov, F., & Yakovlev, V. (2015). The protocols performance evaluation of the data link layer in the iso/osi model (Patent No. RU 2015619739). In*.na.character: Vol.na.character* (Patent RU 2015619739)..na.character. https://doi.org/.na.character

11. Yakovlev, V., & Kushnazarov, F. (2015). Evaluation of the effect of interferences on link-layer protocol performance. *Proceeding of Petersburg Transport University*, *1*(42), 133–138. https://doi.org/.na.character

12. Кушназаров, Ф. И., Яковлев, В. В., & Турдиев, О. А. (2015). Сравнение производительности протоколов доступа к облачным ресурсам. *Известия Петербургского университета путей сообщения*,*.na.character*(4 (45)),.na.character. https://doi.org/.na.character

13. Кушназаров, Ф. И., Турдиев, О. А., & Кушназаров, Ф. И. (2015). ЭФФЕКТИВНОСТЬ циклических кодов при передаче данных в каналах связи с помехами. *Сборник трудов VII научно-практической конференции молодых ученых*,*.na.character*(73),.na.character. https://doi.org/.na.character

## Projects

### NLP: Speech Accent Detection

*Project Link*

*2020*

Everyone who speaks a language, speaks it with an accent. This project defines accents for the english language speakers

- Role: Author and Maintainer
- Results: Accuracy=.90, Recall=.91 and Precision=.93
- Increase accuracy of ASR (Automatic Speech Recognition)

### NLP: Text classification

*Private*

*2019*

Classified to 10 different topics

- Data sources: Title, Description, Text
- Method: LDA
- Result: 10 topics. Accuracy=.87

### Moscow Smart Lighting Projects

*Project Link*

*2015*

Moscow Smart Lighting Projects Project, which is made by 5 different office building groups with different requirements to smart system and linking. Companies main job is to make computer programming of center control and night light adjustment. More then 800 square meters, with 15000 signals from different controllers

- Maintainer

**L-Cube** *Company WebPage*

Implemented at the company L-Cube LLC in order to optimize the performance of the departmental network *2015*

- Author and Maintainer

**EMC-DELL** *Implementation Certificate*

Certificates from EMC2 Data-Center about Value of Research and Possibility of Implementation the Model *2016*

- Author and Maintainer

**IntellektTrans** *Conference WebPage*

Intelligent rail system transportation summit. As one of the members of the Organizing Committee of the intelligent rail system transportation summit, the organization participated in the summit and attended the meeting *2009*

- Volunteer

## Accomplishments

| | | |
|---|---|---|
| 2020 | IBM Data Science | *IBM and Coursera* |
| 2020 | Applied Data Science Capstone | *IBM and Coursera* |
| 2020 | Machine Learning with Python | *IBM and Coursera* |
| 2020 | Data Analysis with Python | *IBM and Coursera* |
| 2020 | Databases and SQL for Data Science | *IBM and Coursera* |
| 2020 | Python for Data Science and AI | *IBM and Coursera* |
| 2020 | Data Science Methodology | *IBM and Coursera* |
| 2020 | Tools for Data Science | *IBM and Coursera* |
| 2020 | Data Visualization with Python | *IBM and Coursera* |
| 2020 | What's Data Scientists | *IBM and Coursera* |
| 2017 | Big Data 101 | *Big Data University* |
| 2016 | Data Mining | *Intuit National Open University* |
| 2013 | Certificate of knowledge of technology AMX I - II level | *Smart City and IoT* |
| 2013 | Certificate of knowledge of ABB technology I-III | *Smart City and IoT* |
| 2013 | Certificate knowledge of technology Crestron I - II level | *Smart City and IoT* |
| 2012 | Data Science - effective use of data | *DELL-EMC* |
| 2012 | ISM - administration of information and data storage | *DELL-EMC* |
| 2012 | The C Programming Language | *Intuit National Open University* |
| 2007 | Fundamentals of IBM z Series | *IBM* |